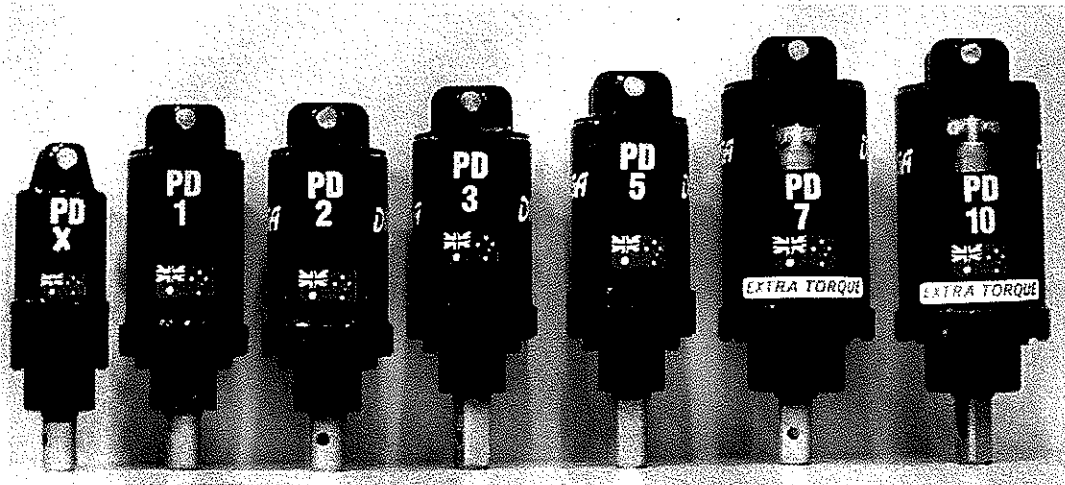


DIGGA

Operators/Parts

Manual



EARTH AUGER DRIVE UNITS

PD MODELS – TO SUIT THE FOLLOWING MACHINES

[] PDX

Mini Excavators (800 kg to 1.5 Tonnes) – Small Truck Cranes – Mini Skid Steer Loaders

[] PD1

Skid Steer Loaders – Excavators (1 to 5 Tonnes) – Mini Backhoes – Truck Cranes – Tractor Loaders

[] PD2

[] PD3

Skid Steer Loaders – Backhoes – Excavators (3 to 7 Tonnes) – Truck Cranes – Telescopic Handlers

[] PD5

[] PD7

Excavators (5 to 10 Tonnes) – Backhoes – Telescopic Handlers – Tool Carriers – Truck Cranes

[] PD10

DISTRIBUTED BY

**DIGGA EUROPE Ltd
PO BOX 39
CHELTENHAM
GLOUCESTERSHIRE GL53 7ZA
UNITED KINGDOM**

MANUFACTURED BY

**DIGGA AUSTRALIA Pty Ltd
25 CHRISTENSEN ROAD
STAPYLTON
QUEENSLAND
AUSTRALIA 4207**

DIGGA

Manufacturers of Planetary Drives - Trenchers - Augers - Wearparts



CONTENTS

To the Purchaser	2
Operator & General Safety	3
Preparation Before Set-up	4
Set Up Instructions	5
Operating Instructions	6
Description Diagram	7
PDX Exploded View	8
PDX Parts List	9
PD1/PD2/PD3/PD5 Exploded View	10
PD1/PD2/PD3/PD5 Parts List	11
PD7/PD10 Exploded View	12
PD7/PD10 Parts List	13
Lubrication	14
Troubleshooting	15
Augers To Suit A4 & A5 Euro range	16
Wearparts To Suit A4 & A5 Augers	17
Warranty Statement	18
Risk Assessment Acknowledgement	19
Risk Assessment – Skid Steer & Excavator	

Thank you for purchasing your new **DIGGA** product.

DIGGA products are designed and manufactured to give dependable service, to keep them running efficiently, and for safety reasons **ENSURE ALL OPERATORS READ** this **OPERATORS/PARTS MANUAL** and the **RISK MANAGEMENT STRATEGIES MANUAL**. Should you require extra manuals please contact your **DIGGA DEALER**.

Optional extras are available for special applications or extreme conditions; these are noted throughout each manual.

Contact your **DIGGA** Dealer for any further information or spare parts you may require.

Record your **DIGGA Product** details in the spaces below.

Your **DIGGA** Dealer will need these details to give you prompt, efficient service.

DIGGA	AVANT GF35		Date of Manufacture	21.1.99
Serial No.	UK	CE	Manufactured by DIGGA Australia Pty Ltd. 25 Christensen Rd Stapylton, Qld. Australia 4207.	
Flow	MAX: 75	LPM	Distributed by DIGGA Europe Ltd. P.O. Box 39 Cheltenham Glos GL 53 7ZA United Kingdom, Tel + 44(0) 8709 010 002	
Pressure	MAX: 205	BAR		
Weight	55kg			

Purchased From:

Date ____/____/____

NOTE: As **DIGGA** continually strives to improve and increase it's range of products, the right to alter specifications at any time without notice or obligation, is reserved.

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Planetary Drive Unit Operators Manual

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The safety of the operator and the safety of everyone within the work area depend on the operator's knowledge of the **DIGGA** Product. Before any operation commences with the **DIGGA** Product ENSURE **ALL OPERATORS READ** this **OPERATORS/PARTS MANUAL** and **RISK MANAGEMENT STRATEGIES MANUAL** and adhere to all safety guidelines.

CAUTION: Complete a daily inspection prior to operating any **DIGGA** product.

CAUTION: Replace worn parts. Worn parts may endanger safety.

CAUTION: Wear close fitting protective clothing and correct safety footwear.

CAUTION: Always ensure onlookers are at least six (6) metres from all moving parts.

CAUTION: Keep all hands, feet and clothing well away from all moving parts.

CAUTION: Be alert at all times while operating the **DIGGA** product and always ensure you are in complete control.

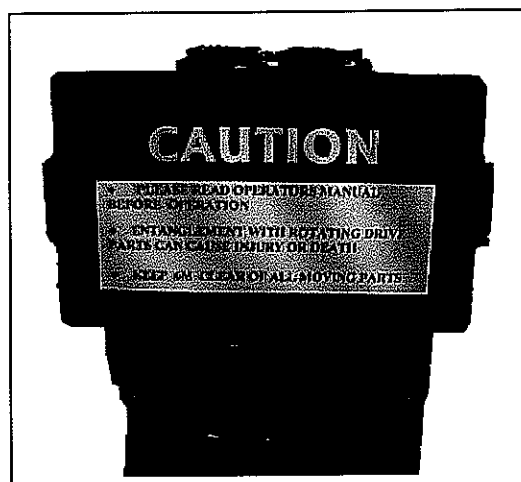
CAUTION: Always be aware of any obstructions that may cause injury to the operator or damage the **DIGGA** product.

CAUTION: Be aware at all times of any Warning, Caution or other signs within the working area.

NOTE: It is your responsibility as the operator to check for all underground services such as Gas lines, Electricity, Water lines, Telephone lines, Cable Television etc. before Drilling or Trenching.



Alert yourself to the weight(s) of the DIGGA Product(s) you



Typical Caution Sticker

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PREPARATION BEFORE SETUP

4

This **OPERATORS/PARTS MANUAL** covers the following **DIGGA** product.

AUGER DRIVE UNITS

PDX	PD1	PD2	PD3
PD5	PD7	PD10	

AUGERS

A4	A5 (Euro)
-----------	------------------

WEARPARTS

TS-1	TS-2	TS-3
TS-C-1	TS-C-2	TS-C-3
PM-1	PM-2	PM-3

CHECKLIST

To avoid any inconvenience before operation, please check that you have received the Items (covered in this manual) which you may have ordered.

EXAMPLE

- 1 x Auger Drive Unit
- 1 x Linkage
- 2 x 30mm Pins complete with 2 x Lynch pins.
- 1 x Auger Pin with 1 x Lynch pin.
- 2 x Hydraulic Hoses.

- 1 x Set of Quick release Couplers.
- 1 x Frame (For mounting Auger Drive Unit machine)

Augers
Wearparts

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Fig 1

The **DIGGA** Earth Auger Drive Unit receives its Hydraulic oil flow and pressure from the Parent Machine through the Auxiliary Hydraulic (Attachment) Circuit via two Quick Release Couplers normally located near the end of the Loader arms, Excavator Dipper or Truck Crane Booms etc. (See Fig 1, typical for excavators).



Fig 3

Mounting Frames are attached to the Bucket Attaching Points on Loader arms and Excavators. On Truck Cranes a Linkage is normally located between the Hook Pin and the **DIGGA** Earth Auger Drive Unit.

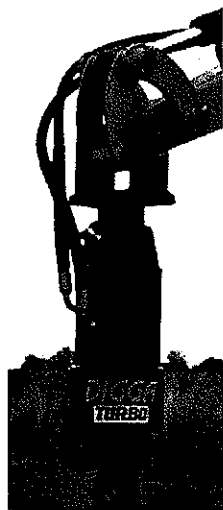


Fig 2

FIG 2: Typical mounting for Excavators.

FIG 3: Typical mounting for Skid Steers.

Fig 4

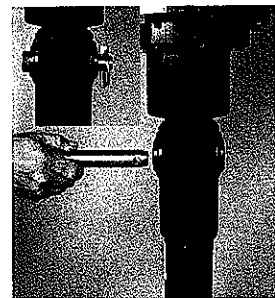


FIG 4: **DIGGA** Auger Fitting



CAUTION: Alert yourself to the weight(s) of the **DIGGA** product(s) You operate. Do not exceed your machine operating weight or lifting capacity. (Refer to your machine manual)

Ensure your machines Flow & Pressures do not exceed the maximum rated Flow & Pressures of the **DIGGA** Earth Auger Drive Unit. (Refer Pg 2 **To the Purchaser**)

- (1) Fit Mounting Frame (Refer Above)
- (2) Fit **DIGGA Earth Auger Drive Unit** using Linkage and Pins.
- (3) Hook up hoses to Auxiliary Hydraulic Circuit. **ENGINE MUST BE SHUT OFF.**
- (4) Check that the drilling rotation on **DIGGA Earth Auger Drive Unit** is clockwise.
NOTE: Two-way flow is required for anticlockwise rotation.
- (5) To assure optimum Hydraulic Motor life, run new **DIGGA Earth Auger Drive Unit** for one hour on approx. 1/3rd RPM before application of full load.
- (6) Fit Auger. (See Fig 4) Use only genuine **DIGGA** Auger Pin and Lynch Pin.

REMOVAL PROCEDURE: Reversal of Set up Instructions.

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OPERATING INSTRUCTIONS

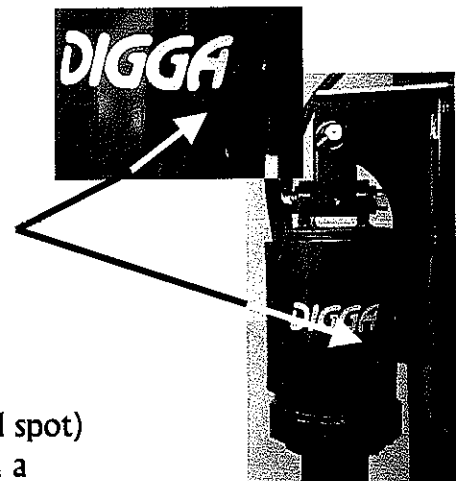
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Before operating, always ensure that the **DIGGA** Earth Auger Drive Unit and Auger are fitted correctly to the parent machine. (Refer to Set up Instructions Pg 5).

CAUTION: Do not operate on excessive slopes, embankments or near edges of excavations.

When travelling or manoeuvring Parent Machine to the required drilling area, reduce ground speed to prevent the Earth Auger Drive Unit and Auger from swinging.

NOTE: Some mounting Frames are fitted with a support bracket. When fitted ensure support bracket locates the Earth Auger Drive Unit and Auger during travelling or manoeuvring (See Photo, Typical for Skid Steer Loaders)



To commence drilling, locate Auger Pilot (on pre marked spot) and engage machine auxiliary lever to rotate the Auger in a clockwise direction.

Lower the machine arm(s) to push the Auger into the ground. The harder the ground the more down pressure you will need.

NOTE: Do not continually stall the Earth Auger Drive Unit. Continual stalling may cause excessive heating of the hydraulic system.

As the Auger digs into the ground , the machine arm(s) may travel through an arc. You may therefore have to continually reposition the Earth Auger Drive Unit or Parent machine to drill vertically.

Do not drill the Auger deep into the ground in one operation. Continually clearing the hole during drilling will facilitate a faster and easier operation.

NOTE: Extension shafts are available for extra depth.

Disengage machine auxiliary lever and rest Earth Auger Drive Unit and Auger on the ground before making any adjustments or leaving the Parent Machine.

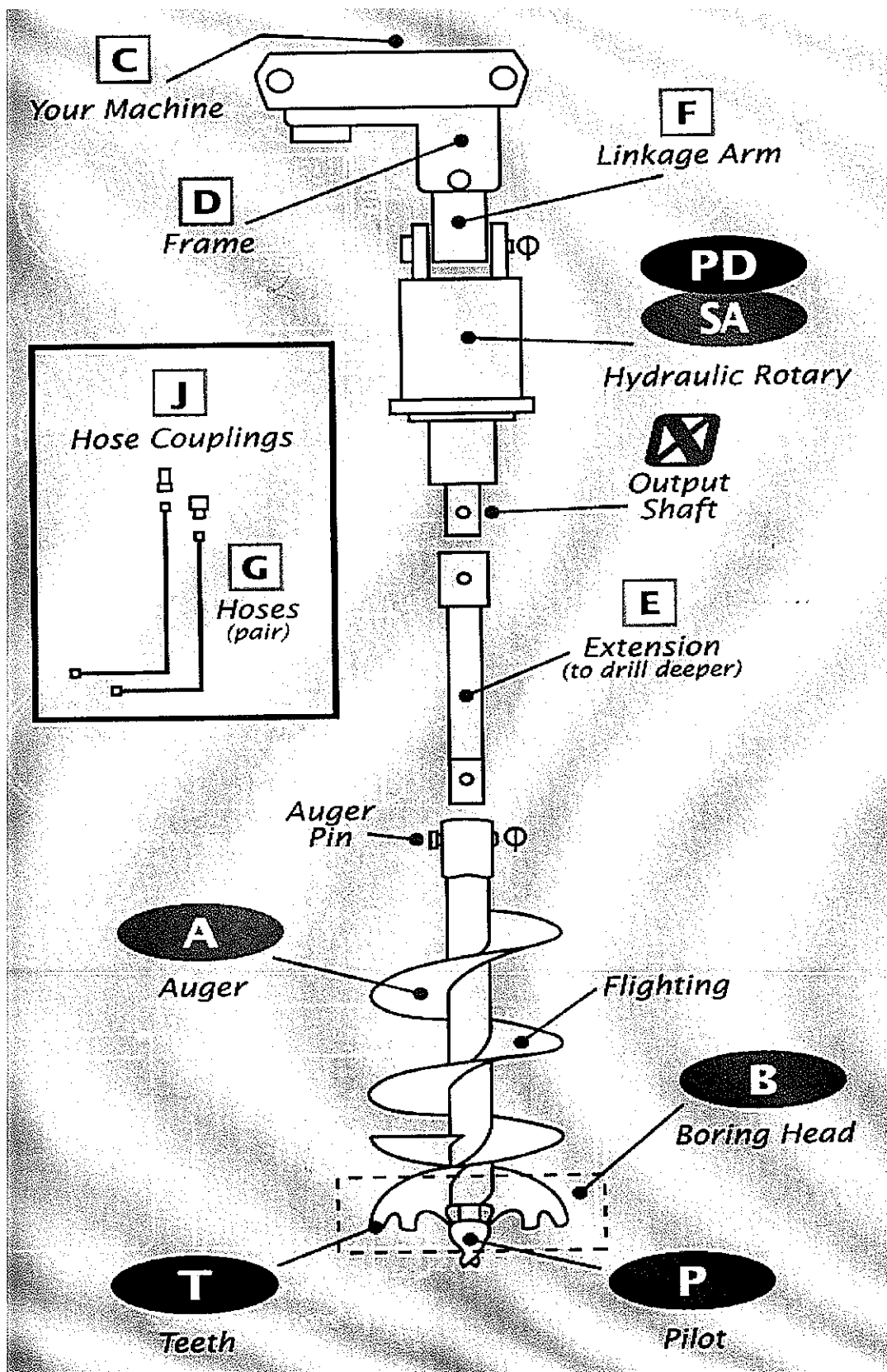
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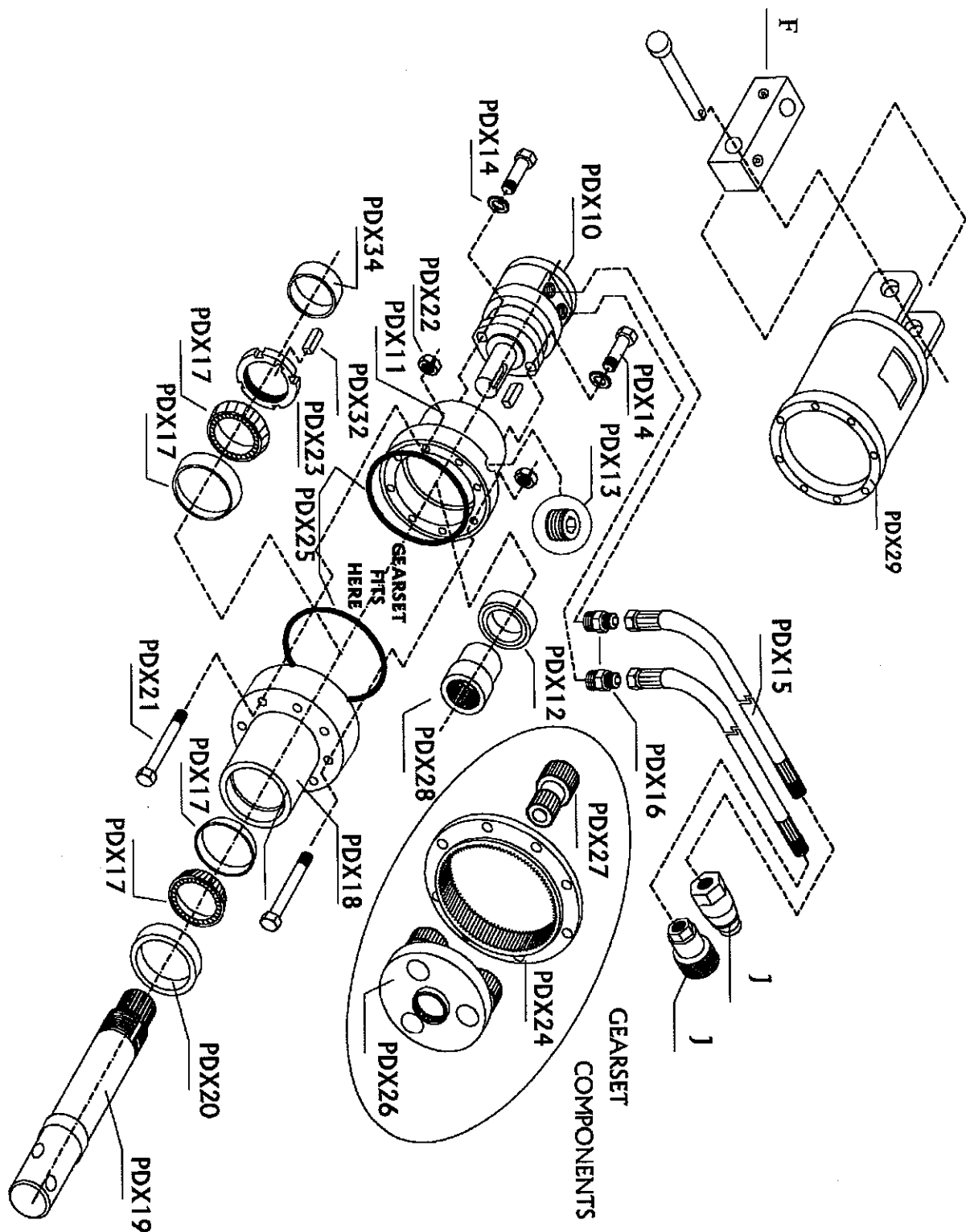
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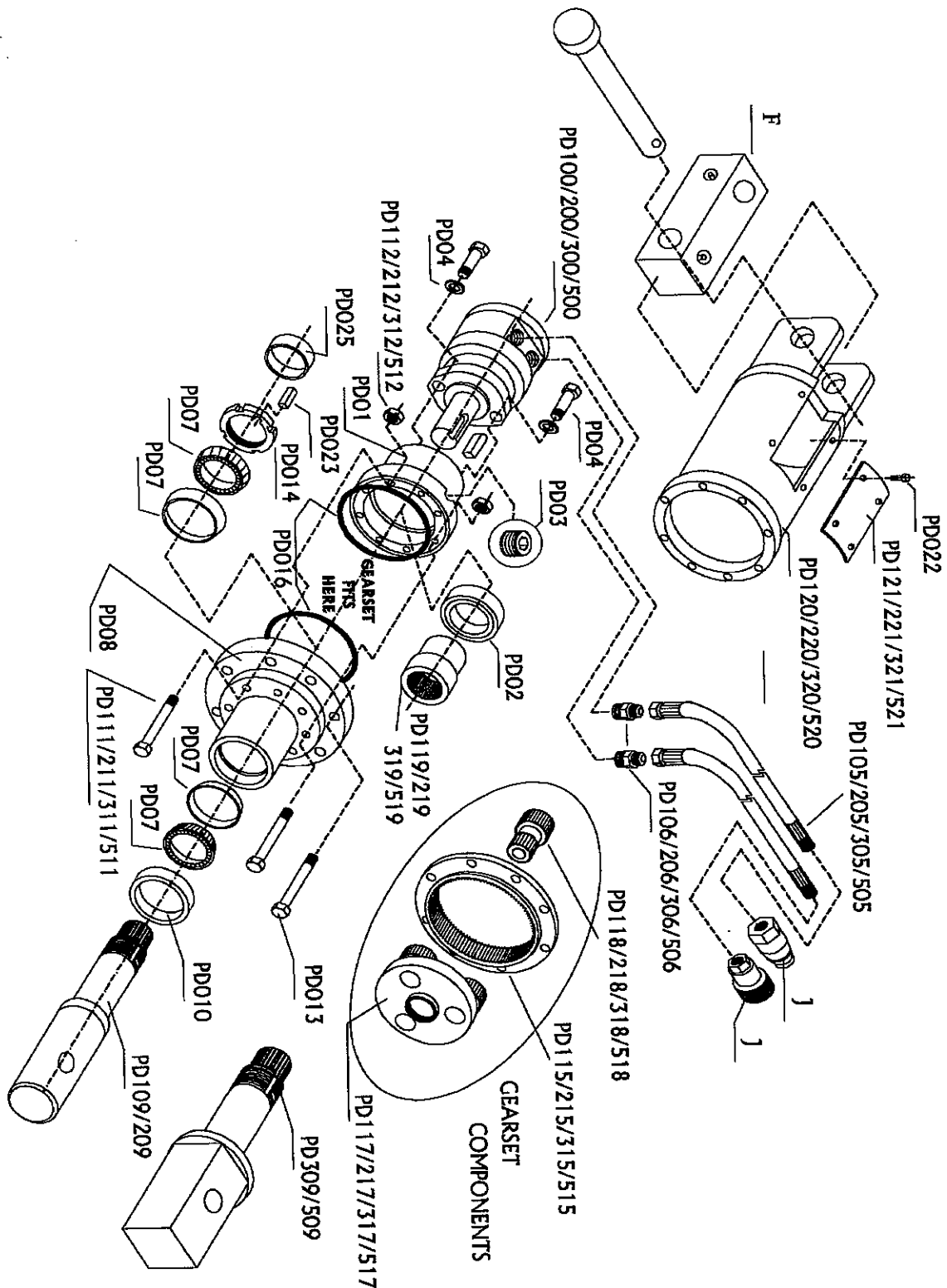




Part No.	Description	No. Per Unit
PDX 10	Hydraulic Motor	1
PDX 11	Input Housing	1
PDX 12	Input Bearing	1
PDX 13	Filler Plugs	1
PDX 14	½" x 1 ¼" UNC H.T Bolt & Spring Washer	2
PDX 15	Hose 2.2m Long	1 Pair
PDX 16	7/8" UNO x ¾" JIC Male-Male Nipple	2
PDX 17	Tapered Thrust Bearing	2
PDX 18	Output Housing	1
PDX 19	Output Shaft (Various options)	1
PDX 20	Oil Seal	1
PDX 21	Gearbox Bolt	8
PDX 22	Nyloc Nut (Suit Gearbox Bolt)	8
	Hood Retaining Bolts (If Applicable)	8
PDX 23	Locking Nut	1
PDX 24	Ring Gear	1
PDX 25	'O' Rings	2
PDX 26	Carrier	1
PDX 27	Input Pinion	1
PDX 28	Input Adaptor	1
	Drive Hood	1
	Drive Hood Cover	1
	Drive Hood Cover Bolts	4
PDX 32	Locking Key	1
PDX 33	Gear Oil	450ml
PDX 34	Spacer Ring	1
F	Linkage	1

OPTIONS: (See Description Diagram Pg7)

A	Auger with	B	T	and	P
D	Frame				
E	Extension				
J	Hose Couplings Q.R.C's				



PD1/PD2/PD3/PD5 PARTS LIST

11

P/No PD1	P/No PD2	P/No PD3	P/No PD5	Description	No. Per Unit
PD100	PD200	PD300	PD500	Hydraulic Motor	1
		PD01		Input Housing	1
		PD02		Input Bearing	1
		PD03		Filler Plugs	1
		PD04		1/2" x 1 1/4" UNC H.T Bolt & Spring Washer	2
PD105	PD205	PD305	PD505	Hose 2.2m Long	1 Pair
PD106	PD206	PD306	PD506	7/8" UNO x 3/4" JIC Male-Male Nipple	2
		PD07		Tapered Thrust Bearing	2
		PD08		Output Housing	1
PD109	PD209			Output Shaft std 65mm Rnd (Various options)	1
		PD309	PD509	Output Shaft std 57mm Sq (Various options)	
		PD010		Oil Seal	1
PD111	PD211	PD311	PD511	Gearbox Bolt	8
PD112	PD212	PD312	PD512	Nyloc Nut (Suit Gearbox Bolt)	8
		PD013		Hood Retaining Bolts (If Applicable)	8
		PD014		Locking Nut	1
PD115	PD215	PD315	PD515	Ring Gear	1
		PD016		'O' Rings	2
PD117	PD217	PD317	PD517	Carrier	1
		PD118		Input Pinion	1
		PD119		Input Adaptor	1
PD120	PD220	PD320	PD520	Drive Hood	1
PD121	PD221	PD321	PD521	Drive Hood Cover	1
		PD022		Drive Hood Cover Bolts	4
		PD023		Locking Key	1
PD124	PD224			Gear Oil	600ml
		PD324	PD524	Gear Oil	650ml
		PD025		Spacer Ring	1
		F		Linkage	1

OPTIONS: (See Description Diagram Pg 7) Shaft Options (See Chart Pg 16)

A Auger with **B** **T** and **P**

D Frame **E** Extension **J** Hose Couplings Q.R.C's

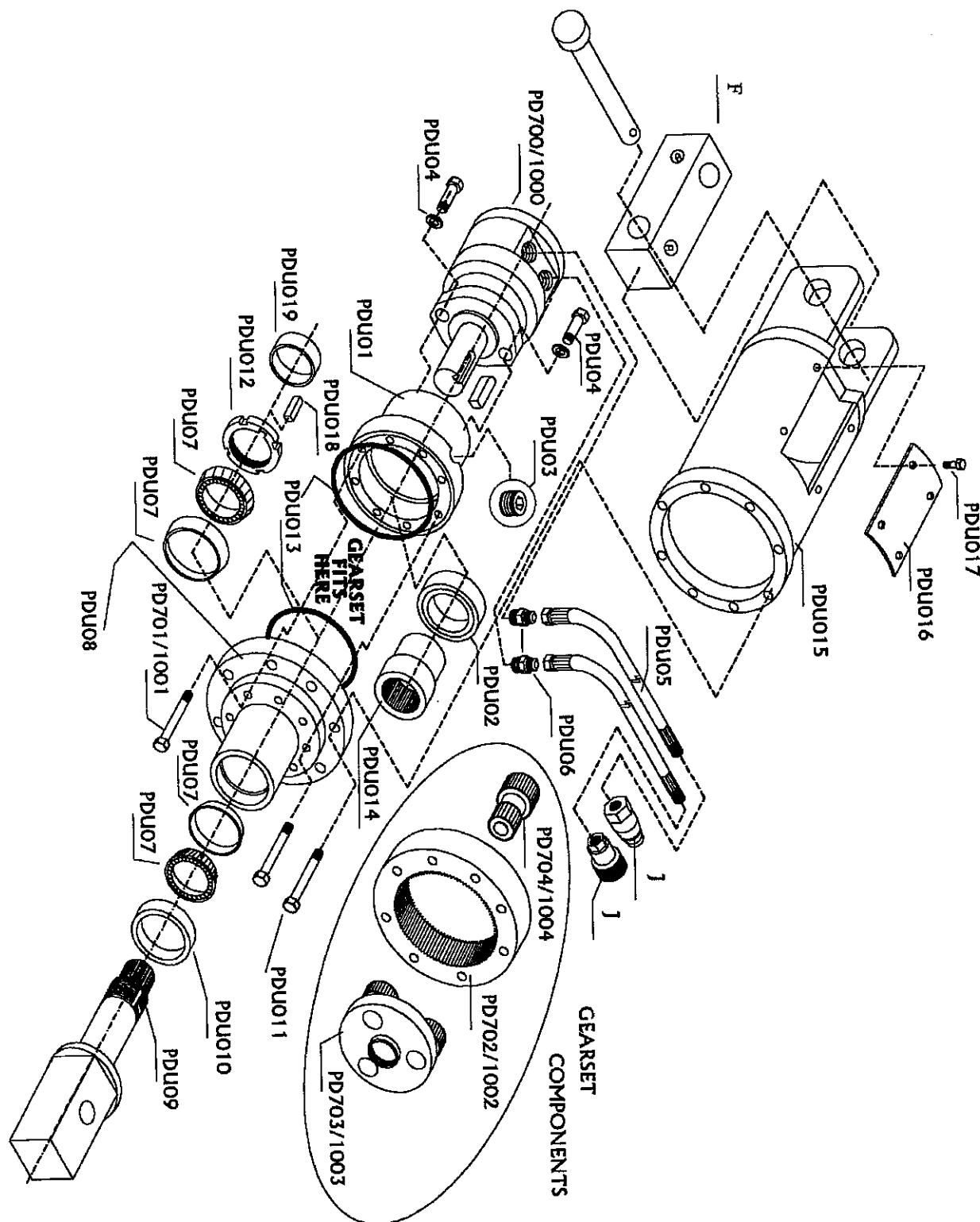
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P/No PD7	P/No PD10	Description	No. Per Unit
PD700	PD1000	Hydraulic Motor	1
PDU01		Input Housing	1
PDU02		Input Bearing	1
PDU03		Filler Plugs	1
PDU04		1/2" x 1 1/4" UNC H.T Bolt & Spring Washer	2
PDU05		Hose 2.2m Long	1 Pair
PDU06		1 1/16" UNO x 7/8" JIC Male-Male Nipple	2
PDU07		Tapered Thrust Bearing	2
PDU08		Output Housing	1
PDU09		Output Shaft Std 57mm Sq (Various options)	1
PDU010		Oil Seal	1
PD701	PD1001	Gearbox Bolt	8
PDU011		Hood Retaining Bolts (If Applicable)	8
PDU012		Locking Nut	1
PD702	PD1002	Ring Gear	1
PDU013		'O' Rings	2
PD703	PD1003	Carrier	1
PD704	PD1004	Input Pinion	1
PDU014		Input Adaptor	1
PDU015		Drive Hood	1
PDU016		Drive Hood Cover	1
PDU017		Drive Hood Cover Bolts	4
PDU018		Locking Key	1
PD705		Gear Oil	ml
	PD1005	Gear Oil	ml
PDU019		Spacer Ring	1
F		Linkage	1

OPTIONS: (See Description Diagram Pg 7) Shaft Options (See Chart Pg 16)

A Auger with **B** **T** and **P**

D Frame **E** Extension **J** Hose Couplings Q.R.C's

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EARTH AUGER DRIVE UNIT

The Earth Auger Drive Unit has sealed gear housing filled with Gear Oil to lubricate the Planetary Gearset Components and Bearings within the housing.

One of the main features of the DIGGA Earth Auger Drive Units is low maintenance, However regular checks for oil leaks and following the service schedules below are recommended to ensure you get a trouble free product.

First Service (Labour Free)

The Gear Oil requires draining and replaced with Castrol SP320 (or equivalent) after the first 500 hours of use or six (6) months from date of purchase – which ever occurs first.

NOTE: Return the Earth Auger Drive Unit to your DIGGA Dealer for the above Labour Free Service.

Yearly or 2000 Hour Service

After the First Service the Gear Oil requires draining and replaced with Castrol SP320 (or equivalent) every twelve (12) months or 2000 hours – which ever occurs first.

NOTE: Should you drain and replace the Gear Oil yourself, FIRST, contact your DIGGA

UNITS	PDX	PD1	PD2	PD3	PD5	PD7	PD10
OIL ml	450	450	450	500	500	600	600

NOTE: Oil should be changed when hot, to prevent build up of sludge deposits. If necessary, flush with fluid (recommended by Oil Companies) to remove any sludge deposits.
Cleanliness is necessary when changing oil.

CAUTION: If you see any sign of an OIL LEAK, inspect and attend to the repair immediately. Your Earth Auger Drive Unit contains only a small amount of Gear Oil. (Refer to the Trouble Shooting Pg 15)

CAUTION: If changing or topping up Gear Oil, use only Castrol SP320 (or equivalent). DO NOT mix oil of different viscosity.
DO NOT mix mineral and synthetic oils.

CAUTION: Continuous operating temperature must not exceed 80° c.

CAUTION During extending storage periods, every month run the Earth Auger Drive Unit to immerse all internals in oil, there by preventing internal corrosion

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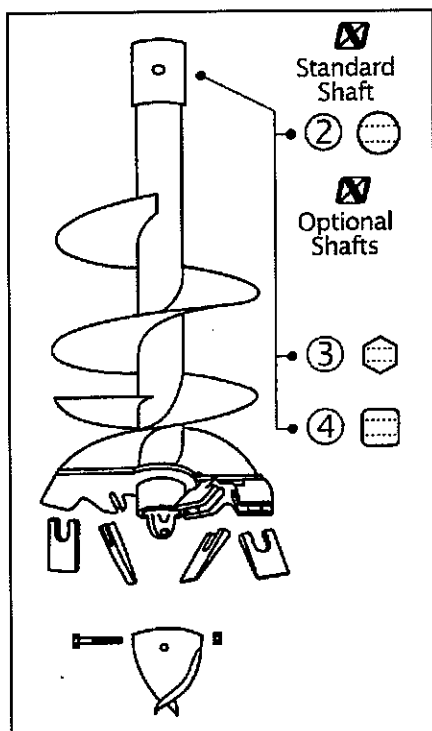
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TROUBLE	POSSIBLE CAUSE	REMEDY
EARTH AUGER DRIVE UNITS		
No Rotation	Quick Release Coupler(s) not Engaged Quick Release Coupler(s) Faulty Auxiliary Valve on Machine Faulty Hydraulic Oil Tank Low Hydraulic Motor Failure Output Shaft Bearing Failure Planetary Gear Failure Machine Oil Pump Faulty	Check Quick Release Coupler(s) Replace Faulty Coupler(s) Refer to Machine Manual Fill Oil Tank to Maximum Level Contact your DIGGA Dealer Contact your DIGGA Dealer Contact your DIGGA Dealer Refer to Machine Manual
Slow Rotation	Low Oil Flow Drive Unit too Large for Machine Hydraulic System too Hot	Check Machine Specifications Contact your DIGGA Dealer See Hydraulic Section (Below)
Hood Leaking Oil	Hose(s) or Fitting(s) Leaking Motor 'O' Ring Failure	Tighten or Replace Contact your DIGGA Dealer
Output Shaft Leaking Oil	Oil Seal Failure Hydraulic Motor Failure	Contact your DIGGA Dealer Contact your DIGGA Dealer
No Torque	Oil Pressure too Low Drive Unit too Small for Machine Hydraulic System too Hot	Check machine Specifications Contact your DIGGA Dealer See Hydraulic Section (Below)
HYDRAULIC SYSTEM		
Oil Over Heating	Oil Pressure too Low Restriction in Line Auger Continually Stalling Drive Unit too Small Machine too Small Hydraulic Oil Tank Low Insufficient Oil Capacity	Set Relief Valve to Machine Spec Inspect and Repair Limit Down Pressure Contact your DIGGA Dealer Fit Drive Unit to Larger Machine Fill Oil Tank to Maximum Level Fit Oil Cooler
AUGERS		
Slow Digging Speed	Worn Teeth or Pilot Ground too Hard Low Oil Flow Auger too Large for Drive Unit Machine too Small	Replace (See Wearparts Pg 17) Contact your DIGGA Dealer Check Machine Specifications Fit Larger Drive Unit Fit Drive Unit to Larger Machine
Continual Stalling	Excessive Down Pressure Oil Pressure too Low Auger too Large for Drive Unit Machine too Small Hydraulic System too Hot	Limit Down Pressure Check Machine Specifications Fit Larger Drive Unit Fit Drive Unit to Larger Machine See Hydraulic Section (Above)
Bent Shaft or Flighting	Abuse Fitted to Wrong Drive Unit	Replace or Repair Refer to Pg 16
WEARPARTS		
Excessive Wear	Abrasive Ground	Contact your DIGGA Dealer



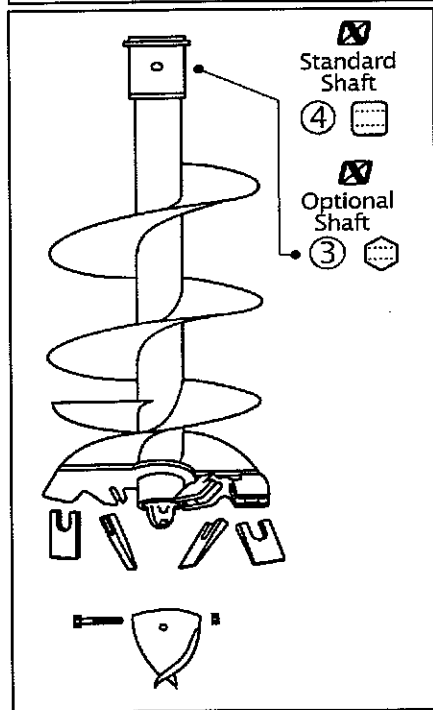
AUGER A4

**A4 TO SUIT:
PDX / PD1 / PD2**

SIZE mm	WEIGHT Kg
100	8.5
150	17.4
200	19.1
225	23.7
250	26.1
300	30.0
350	32.7
400	38.1
450	39.7
500	44.6
600	56.7
750	72.2

SHAFT OPTIONS

②
65mm Round Drive
③
2" Hexagonal Drive
④
57mm Square Drive



AUGER A5 (Euro Range)

**A5 TO SUIT:
PD3 / PD5
PD7 / PD10**

SIZE mm	WEIGHT Kg
150	24.0
200	27.6
225	31.2
250	32.9
300	44.5
350	50.2
400	60.1
450	65.2
500	75.1
600	92.2
750	98.7
900	123.4

AUGER MAINTENANCE

As the Auger is engaging the ground, then wear must occur to dig holes. Therefore, the Auger Teeth & Pilot must be checked regularly and replaced with new wearparts. (See Wearparts Pg 17)

Failure to do so will damage the Auger Pockets and Flighting (See Description Diagram Pg 7) This

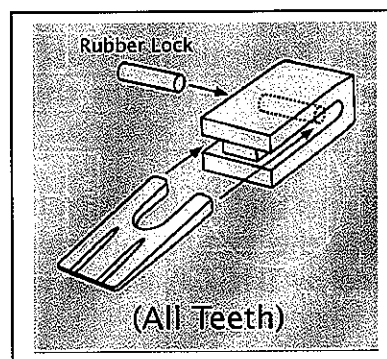
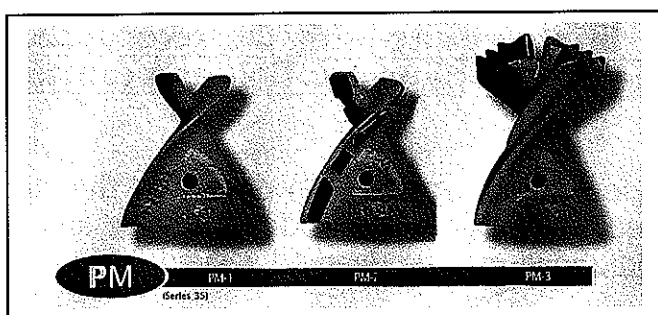
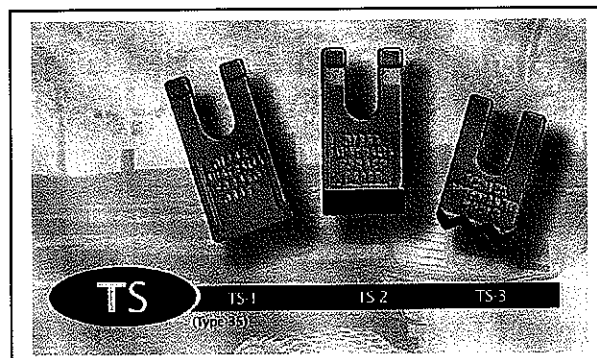
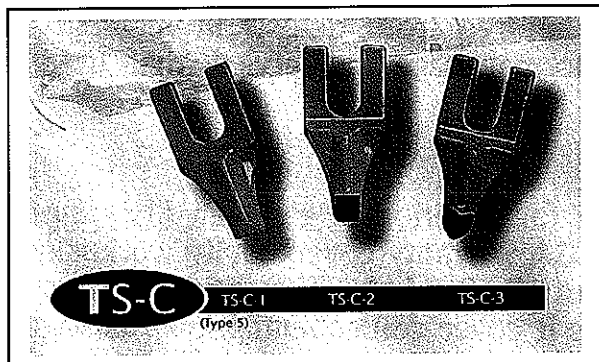
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TEETH REPLACEMENT

CAUTION: Wear Safety Glasses for Following Procedure.

- (1) Drive out worn teeth using a 6mm flat pin punch.
- (2) Remove old Rubber Lock.
- (3) Cut new Rubber Lock to length, 30mm, then fit to pocket.
- (4) Fit New Teeth using Copper Hammer or Soft Headed Hammer.
(Do Not Use A Steel Hammer)

NOTES: (a) TS-C (Chisel Teeth) – Must only be fitted to Inner Pockets.
(b) Tungsten Teeth – Tungsten tips must face upwards.

PILOT REPLACEMENT

- (1) Remove Worn Pilot using suitable spanners.
- (2) Fit New Pilot.

NOTE: (a) Fit correct size Bolt and Nyloc Nut.

All new **DIGGA** products are warranted to be free from defects in material and workmanship, which may cause failure under normal usage and service when used for the purpose, intended.

Digga Australia Pty Ltd warrants its equipment for a period of eighteen (18) months dating from delivery to original user. The **DIGGA** warranty covers faulty workmanship and defective parts manufactured by Digga Australia Pty Ltd. The warranty does not extend to transportation cost of parts nor does it cover consequential loss, damage to Hydraulic Hoses or Ground Engaging parts such as Sprockets, Digging Chain, Bearings and Teeth.

DIGGA Equipment must be operated in accordance with the recommended procedures and within the ranges as specified both on the Unit and contained in the Operating Manual.

Any claims under this warranty must be made within fourteen (14) days after the buyer learns of the facts upon which claim is based. All claims not made in writing and not received by **DIGGA** within the time specified above may be deemed waived.

DIGGA will not be responsible for or accept any charges for work carried out by any repairers, or for charges for any spare parts fitted to any **DIGGA** products without written approval from **DIGGA**.

This warranty is in lieu of all other warranties expressed or implied and there are no warranties or merchant ability or of fitness for a particular purpose. In no event shall **DIGGA** be liable for consequential or special damages.

DIGGA'S liability for any and all losses and damages to buyer resulting from any cause whatsoever, including **DIGGA** negligence irrespective of whether such defects are discoverable or latent, shall in no event exceed the purchase price of the particular parts, with respect to which losses or damages are claimed, or, at the discretion of **DIGGA** the repair or replacement of defective or damaged parts.

Any goods returned to **DIGGA** by the customer under Warranty or repair must have all freight charges prepaid for on the customer's Account.

Digga Australia Pty Ltd

APPENDIX 1

ACKNOWLEDGEMENT

(INSTRUCTIONS ON SIGNING)

**Both of these pages need to be signed by the Customer
AND either the Agent for DIGGA Australia Pty Ltd
Or DIGGA Australia Pty Ltd if bought Direct.**

**The unperforated sheet is the customers Original
AND the Copy (perforated sheet) is removed by either:
The Agent for DIGGA Australia Pty Ltd
Or DIGGA Australia Pty Ltd**

Digga Australia Pty Ltd

RISK ASSESSMENT IN COMPLIANCE WITH NATIONAL OCCUPATIONAL HEALTH AND SAFETY COMMISSION DOCUMENT: 1010 (1994)

This document must be dated and signed to be valid. Any changes made to this document proceeding the date specified and revision number below, must also be dated with a revision number, then sent out to all machine owners so that they can update their documents. The recipient must acknowledge the revision by return of the audit notice.

DATE: _____	ISSUE:- 01
SIGNATURE:- _____	
FULL NAME:- _____	
POSITION:- _____	

ORIGINAL ACKNOWLEDGEMENT

RISK MANAGEMENT STRATEGY AND STANDARD WORK PROCEDURES

The purpose of this document is to:-

Acknowledge that I have received the Risk Management Strategy and Standard Work Procedures as supplied by DIGGA AUSTRALIA Pty Ltd, as its agents, servants and assignees, under standard terms and conditions of sale; that I have read and understood the Standard Work Procedures and that I agree to comply fully with the procedures as standard operating practice.

Accept full responsibility for the training of each of my operators, whether employed or under sub-contract, and agree to make available a copy of the Risk Management Strategy and Standard Work Procedures in order that the operator(s) may acknowledge training and ensure full compliance in the operation of the equipment.

Further that I shall keep a training log showing training for each and every operator detailing date of provision, name and address of operator and duration of training. I shall have each operator acknowledge receipt of training and agreement to comply with Standard Work Procedures. This log is to be provided to DIGGA AUSTRALIA Pty Ltd, as its servant, agent or assignee on request.

Signed for and on behalf of:

Customer: _____

Name: _____

Address: _____

Agent for DIGGA Australia Pty Ltd

Digga Australia Pty Ltd

COPY ACKNOWLEDGEMENT

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DISCLAIMER

This document has been produced using information supplied by the principal and by personal observation. Whilst every effort has been made to address every item in the most complete manner, it is possible that some information was not available at the time, or that circumstances or situations may arise which had not or could not be envisaged. Any circumstance or situation, which comes to light, which has not been dealt with in the document, must be considered as a new identified hazard, and the principles of Risk Management applied accordingly.

1.0. INTRODUCTION

- 1.1. Persons who have an obligation under the Workplace Health and Safety Act 1995 (Queensland), or comparable legislation throughout Australia, have an obligation to prevent reasonably foreseeable harm resulting from or associated with plant. The information given through risk management can guide a person as to the measures that should be taken in order to prevent reasonably foreseeable harm associated with plant. Risk Management is limited by the issue of practicability.
- 1.2. Risk Management is a logical and systematic approach to managing the uncertainty regarding risk. Risk Management is basically about.
 - (a) Identifying the hazards that exist with the plant
 - (b) Assessing the risks associated with those hazards
 - (c) Taking appropriate action to eliminate or control those risks
- 1.3. A hazard is something with the potential to cause injury or disease.
A risk is the probability of a hazard resulting in an injury or disease, together with the seriousness of the injury or disease.

2.0. OBJECTIVES

- 2.1. The objective of this "Risk Assessment" is to protect the health and safety of any person from hazards arising from this plant and systems of work associated with this plant by: -
 - (a) Ensuring that hazards associated with the use of this plant in the workplace are identified and risks to health and safety are assessed and controlled.
 - (b) Eliminating, or where this is not practicable, minimising risks to health and safety arising from this plant.
 - (c) Requiring the provision of relevant information and training.

3.0. HAZARD IDENTIFICATION

- 3.1. A number of methods were applied to identify hazards associated with this plant. These included: -
 - 3.1.1. An on site evaluation of the plant at work in a number of situations. On flat land and on slopes was used to help identify hazards and determine risk.
 - 3.1.2. Consultation with organisations using the plant and discussions with management, workers and with contractors using the plant to also identify hazards and determine risk.
 - 3.1.3. The manufacturer's "Procedures Manual" details methods of work for all applications recommended at present. The procedures will be applied to the hazards identified.
- 3.2. All reasonably foreseeable hazards to health and safety arising from this plant and systems of work associated with this plant will be identified.
 - 3.2.1. Without limiting the generality of clause 3.2, hazards associated with the following factors, so far as they are relevant to the design, manufacture, installation and use of this plant will be identified.
 - 3.2.2. **Suitability of the type of plant for the particular task.**
Auger specifically designed for use on "Skid Steer Machine"

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- 3.2.3. Actual and intended use at the workplace.**
Boring/ sinking holes
- 3.2.4. Environmental conditions and terrain in which plant is used.**
Work sites & conditions governed by "Skid Steer Machines Operations Manual". Eg. maximum working degrees/ slopes "Skid Steer Machines" can work on etc.
- 3.2.5. Foreseeable abnormal situations, misuse and fluctuations of operating conditions.**
Steep and uneven ground, using undue force, not using plant as manufacturer intended, changing machine settings and controls, weather conditions.
- 3.2.6. Potential for injury due to entanglement, crushing, trapping, cutting, stabbing, puncturing, shearing, abrasion, tearing and stretching.**
Loose clothing, safety observer/ off sider standing too close to auger,
Skid Steer machine crushing safety observer/ off sider.
Being trapped between auger and machine when attaching & removing auger.
Auger falling from attachment.
Injury occurring while changing augers
Machine rollover.
Muscular/ skeletal injuries
Hydraulic failure causing lift arms to drop.
- 3.2.7. Generation of hazardous conditions due to pressurised content, electricity, noise, radiation, friction, vibration, fire, explosion, temperature, moisture, vapour, gases, dust, ice, hot or cold parts.**
Burst hydraulic hoses. Underground services (electricity, gas, water, sewerage, telecommunication etc.) striking solid objects, airborne particles generated by auger and wind
- 3.2.8. Failure of the plant, resulting in the loss of contents, loss of load, unintended ejection of workpieces, explosion, fragmentation or collapse of parts.**
Replacing auger's bits and pilot bits.
Attachment plate falling off.
Auger pin braking, resulting in auger falling off
- 3.2.9. Capability of plant to lift and move people, equipment and materials, and suitability of secondary backup systems to support the load.**
Capability of plant to lift and move people, equipment and materials.
- 3.2.10. Control systems, including guarding and communications system.** Altering or tampering with controls or systems. Communications between machine operator & safety observer/ off sider. PTO Shaft guarding.
- 3.2.11. Potential for falling objects and the plant to roll over.**
Auger falling on machine or safety observer/ off sider.
Plant rollover. Lift arms to fall due to hydraulic failure.
- 3.2.12. Suitability of materials used for the plant.**
N/A
- 3.2.13. Suitability and conditions of all accessories.**
Incorrect or unsuitable auger bits used.
Inappropriate or unsuitable auger connections and parts.
Wear factors.
Correct plumbing of machine.
Incorrect and/or nil maintenance of "Skid Steer Machine".
- 3.2.14. Ergonomic need relating to installation and use.**
Lifting techniques
- 3.2.15. Carrying out the work without the plant.**
Considered more hazardous to muscular/ skeletal system, more labour intensive and time consuming
- 3.2.16. Location in the workplace and the impact on workplace design and layout** N/A

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- 3.2.17. Suitability and stability of the plant and supports.**
Suitability and support of the plant
- 3.2.18. Presence of persons and other plant in the vicinity.**
Persons and other plant in the vicinity
- 3.2.19. Potential for inadvertent movement or operation of the plant.**
Inadvertent movement or operation of the plant & auger when moving within the work site.
- 3.2.20. Access and egress.**
Access and egress to the work site
- 3.2.21. Competency of operators.**
Operator competency
- 4.1.** The hazards identified with the use and operation of the plant were assessed for risk when used in conjunction with the manufacturer "Procedure Manual".
- 4.2.1.** The following have been taken from the hazard identification
- 4.2.2 Suitability of the type of plant for the particular task.**
The augers have been specifically designed to operate on "Skid Steer Machines".
- 4.2.3. Actual and intended use at the workplace.**
To sink/ auger holes in the ground. Utilising a "Skid Steer Machine" as a power pack and a mode of transport.
- 4.2.4. Environmental condition and terrain in which plant is used.**
To work on work sites that do not exceed the safe working limits of the "Skid Steer Machine" as per the "Skid Steer Machine" "Operations Manual".
- 4.2.5. Foreseeable abnormal situations, misuse and fluctuation of operating condition.**
"Skid Steer Machine" to be worked in accordance to its "Operations Manual".
Using undue down force can destabilise the "Skid Steer Machine" causing it to become unbalanced.
Changing machine settings could cause auger to rotate too fast, causing projectiles etc.
- 4.2.6. Potential for injury due to entanglement, crushing and trapping.**
Do not wear loose clothing.
If a safety observer/ offsider is being used, he shall be in charge of work place, also the operator cannot move the "Skid Steer Machine" unless he has been directed by safety observer/ offsider.
Safety observer/ offsider must only direct the operator to move machine when the safety observer/ offsider deems himself to be in a safe place. Only the operator can attach or detach auger assembly (quick hitch system) from the "Skid Steer Machine". No person shall position themselves directly under the lift arms or auger assembly. One must use correct lift & posture techniques while changing auger or extensions. Engine must be "shut down", controls tagged & locked and machine to be in "rest position" with handbrake applied, before any maintenance can be performed. The operator must not egress the machine whilst the machine is in the "operation mode" or "engine running".
- 4.2.7. Generation of hazardous conditions due to pressurised content, electricity, noise, radiation, friction, vibration, fire, explosion, temperature, moisture, vapour, gases, dust, ice, hot or cold parts.**
Hydraulic hoses must be checked on a daily basis and a site specific Risk Assessment must be undertaken for underground services and other work site hazards, use the appropriate personnel protection equipment (eye protection, hearing protection, (if noise levels exceed levels described by current statutes, regulations, codes or advisory standards), UV protection, Industrial work wear, safety helmets).
- 4.2.8. Failure of the plant resulting in loss of load or unintended ejection of work pieces.**
Use soft-headed hammers for replacing auger bits and pilot bits.
Before commencing work ensure locking handle for quick-hitch is in locked position and secured.
Daily check of auger locking pin.

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4.2.9. Capability of plant to lift and move people, equipment and materials, suitability of secondary back-up systems to support the load.

No capability of plant to lift and move people or equipment or materials.

4.2.10. Control systems including guarding & communication system

No altering or tampering with controls or machine systems. If using a safety observer/ offsider they must be in communication with machine operator at all times. Machine operator can only move machinery on work site at the direction of safety observer/ offsider.

4.2.11. Potential for falling objects and the plant to roll over.

Daily checks of auger locking pin & quick attachment locking handle, being locked & secured.

Daily check of hydraulic hoses & fittings.

If using a safety observer/ offsider, they must inform the machine operator of uneven ground.

"Skid Steer Machine" safety manual and operations manual must be adhered to at all times. Eg. Do not exceed recommended slope/ level the machine can safely operate too.

4.2.12. Suitability of materials used for the plant.

N/A

4.2.13. Suitability and conditions of all accessories.

All augers, connections and parts must be to the recommendations of the manufacturer, or have the authority by the manufacturer to use the other attachments.

All worn and replacement wear parts must be replaced in accordance to the manufacturer recommendations. "Skid Steer Machine" must be maintained in accordance with the provisions with the "Workplace Health & Safety Act", regulations, codes of practice and advisory standards.

4.2.14. Ergonomic needs, relating to installation & use.

Use the correct lifting techniques and postures for manual handling when operating machine & changing augers and extensions.

4.2.15. Carrying out the work without the plant.

The purpose of the plant reduces risk of injury to the muscular/ skeleton systems by reducing manual labour, and also increase productivity.

4.2.16. Location in the work place & the impact on work place design & layout.

N/A

4.2.17. Suitability & stability of the plant & supports.

Undue force must not be put on the machine to lift that machine from the ground resulting in instability, machine must be operated on an even surface. The attachment was designed to work within the parameters of the design of the "Skid Steer Machine".

4.2.18. Presence of persons & other plant in the vicinity.

No person to come within 6 metres of operating plant, without first gaining the attention of the operator and/or safety observer/ off sider and the approval of the operator and/or the safety observer/ offsider to enter the work site. Any excavations or holes that are hazardous or are in a high traffic or public area shall be barricaded &/ or clearly identified & they shall comply with the regulations and provisions of the "Workplace Health & Safety Act", regulations, codes of practice and advisory standards.

4.2.19. Potential for inadvertent movement or operation of the plant.

If using a safety observer/ offsider they must direct the operator of the "Skid Steer Machine" when moving within the work site. The safety observer/ offsider must keep all other persons & plant 6 metres clear of moving "Skid Steer Machine". If one-man operations extra care should be taken in identifying the presence of personnel and plant before and during movement of the machine still maintaining a 6 metre clearance. When travelling with the auger in vertical position, the auger must be kept as close to the ground as possible without damaging the auger. When travelling with auger in a locked horizontal position, the tip of the auger should be marked with a high visibility device, so as to be clearly visible to all other persons on work site. The machine must still be directed by safety observer/ offsider and all plant & persons kept 6 metres clear of moving machine.

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4.2.20. Access & Egress

All access and egress must be done in a safe manner to all persons & plant.

4.2.21. Competency of Operators.

All operators must have all necessary tickets & competency required under the statues, regulations, code of practices & advisory standards, to operate "Skid Steer Machine".

The "Skid Steer Machine" operator must have read and understood the "Operations Manual" of the "Skid Steer Machine".

All operators and safety observer/ offsider must have read & understood the "Operators Manual" & "Standard Work Procedures" & any required training as set out by the manufacturer before using the machine.

5.0 CONTROL MEASURES

Control measures adopted for the operation of the "Skid Steer Machine with Auger Attachment" rely largely on the adoption & separation of other persons on the work site & the general public, by distance.

Another control measure is: if a safety observer/ off sider is used, they shall be in control of the work site and the machine operator, as the safety observer/ off sider has basically an unobstructed view of the work site. "Standard Work Procedures" should be firmly adhered too as well as the manufacturers "Operators Manual"; regular maintenance must be carried out according to the manufacturer specifications.

6.0. SUMMARY

The application of Risk Assessment procedures in accordance with N.O.H.S.C: 1010 would show that provided the procedures documented for the use of the plant are followed meticulously, the hazards identified as associated with its use can be minimised. This Risk Assessment has been applied using information supplied by the manufacturer and by observation. Any alteration in the application of the plant or change in operating conditions will necessitate a reappraisal of the Risk Assessment.

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APPENDIX 2

STANDARD WORK PROCEDURE

DIGGA AUSTRALIA PTY LTD

AUGER ATTACHMENTS

FOR

SKID STEER MACHINE

1.0. SCOPE

- 1.1. This specification covers the sinking; Augering of holes using the "Skid Steer Machine with Auger Attachment".
- 1.2. The above may be carried out in public land, private property, construction & building sites, rural & urban areas & mine sites.
Where conditions suit the use of such a machine

2.0. OBJECTIVES

- 2.1. The objectives of this "Standard Work Procedure" are to follow safe and approved techniques when sinking/ Augering holes.
- 2.2. It is a requirement of the Workplace Health and Safety Act 1995 that safe systems of work are employed and the contents of this "Standard Work Procedure" are a fundamental requirement of the obligations imposed under this Act.
- 2.3. The principal / operator and all relevant staff are responsible for ensuring that defined standards and work practices, described in this and any other relevant "Standard Work Procedure", are applied at the work place and that individual actions do not endanger the health and safety of themselves and others.
- 2.4. The principal and operator are responsible for ascertaining and obtaining all required permits, consents, licenses or other necessary approvals to carry out the work prior to commencement of the work. Any rules or conditions or requirements of any consent, permit or license must be complied with by the principal /operator in full.
- 2.5. At all times during the work, all property such as buildings, fences, equipment, trees or shrubs that are to be retained, must be safe guarded.

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3.0. PUBLIC RELATIONS/ DISCLAIMER

- 3.1. The principal / operator shall take due note of the possibility of adverse public comments and potential harm that can be done to the company's or the customer's image and is advised of the high priority that is placed on retaining this image.
- 3.2. Any complaints from the public are to be treated courteously and these, as well as any media enquiries shall be reported to the company and the customer's supervising officer, as soon as possible.
- 3.3. In all instances hole sinking/ Augering work may only be carried out where this work has received the required documented approvals from all state and / or local government authorities, public organisations or private owners. Particular care should be taken where under ground services may be located.
- 3.4. These "Standard Work Procedures" are a guide only and it is the Principal / Operator's responsibility to ascertain and comply with detailed work procedures appropriate for each particular circumstance.
- 3.5. "Digga Australia Pty.Ltd." makes no representation or warranty whatsoever as to whether the above guidelines are comprehensive and the Principal and operator acknowledge that by the use of these guidelines they take full responsibility for any damage, consequential loss or other action or claim suffered or incurred by the Principal or Operator or any third party as a result of operating the "Skid Steer Machine with Auger Attachment" in accordance with these guidelines.
- 3.7. The Principal and Operator free and release "Digga Australia Pty. Ltd." from all liability for any loss suffered or claim or action made against the Principal or Operator during or as a result of operating the "Skid Steer Machine with Auger Attachment" following these work procedures.

4.0. PLANT, TOOLS, EQUIPMENT AND MAINTENANCE

4.1. PLANT

- 4.1.1. In the event that the work requires the use of other plant, such as excavators, front end loaders or trucks, in a supportive role, the Principal Contractor must ensure that this supportive plant is of a standard and is operated in accordance with approved "Standard Work Procedures", government requirements, manufacturers "Operators Manuals" etc.

4.1.2.

4.2. TOOLS AND EQUIPMENT

- 4.2.1. Appropriate safety clothing, i.e. safety helmet, safety boots etc. in accordance with the Workplace Health and Safety Act and relevant industry standard safety requirements.

- 4.2.2. Roadway warning signs in accordance with the "Queensland Transport Manual of Uniform Traffic Control Devices; Part 3-Works on Roads, or equivalent.

4.3. MAINTENANCE

- 4.3.1 All plant, tools and equipment are to be maintained in good working order in conformance with the requirements of the relevant Acts, Regulations or Ordinances.
- 4.3.2. During any periodic maintenance on the "Skid Steer Machine with Auger Attachment" or other attachments, it is required that the "Skid Steer Machine" be positioned for maintenance and the engine of the machine then is stopped. All controls must be tagged and locked (with the key removed) in the "rest position" and all pressure released from the system.

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- 4.3.3.** At any time that the machine is under maintenance, maintenance personnel must use a lockout system to eliminate the possibility of accidental start up. I.e.: All controls tagged and locked, with the maintenance personnel retaining the key until the maintenance period is finished.

5.0. SAFETY

- 5.1.** An "Operating Procedures Risk Assessment" in accordance with the requirements of the Workplace Health and Safety Act 1995 has been carried out.
- 5.2.** All work must be carried out in accordance with the Risk Assessment and with existing government and / or relevant industry safety requirements.
- 5.3.** All work must be carried out in accordance with the Manufacturer's "Owner's Manual"
- 5.4.** The "Skid Steer Operation Manual" must have been read and understood by the operator.
- 5.5.** The operator and safety observer/ off sider both must have read and understood the Digga Australia Pty Ltd "Operation Manual", "Risk Assessments" & "Standard Work Procedures" before any work is commenced.
- 5.6.** Any unauthorised modification carried out to the machine voids any Risk Assessment, warranty and responsibility of the manufacturers and their agents.
- 5.7.** Regardless of the type of work to be carried out and before any work is commenced, the operator must be aware and inform himself of any potential hazards that might exist and take appropriate measures to eliminate them.
- 5.8.** The operator shall especially be aware of the hazards of working near underground services and shall instruct his staff in such matters as detailed in existing industry safety specifications.
- 5.9.** When accessing the machine or refueling the machine the operator shall stop the engine and bring all hydraulic controls to the "rest position".

6.0. ACCREDITATION AND TRAINING

All operators must have read and understood the manufacturers "Operators Manual" and "Standard Work Procedures" and required to adhere too and/or perform the following: -

- 6.1** Adequate training by an approved person.
- 6.2.** Any training conducted by Digga Australia Pty Ltd or their nominated agents must be documented on completion.
- 6.3.** Any further training conducted by the owner must also be documented to identify trained operators.
- 6.4.** Any other accreditation, training and / or licensing to meet the specific requirements of any other relevant industry, e.g. unrestricted ACDC license, safety awareness for non-electrical workers, first aid and site induction courses etc.

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7.0. WORK PROCEDURES

All operators and/or safety observers/offsiders must have read and understood The "Operators Manual", "Standard Work Procedure" and performed any Required training as set out by the manufacturer before using this machine.

- 7.1. The operator of the "Skid Steer Machine" must have all necessary tickets and competency required under the statues, regulations, codes of practice and advisory standards, to operate a "Skid Steer Machine".
- 7.2. Before work commences, ensure that all safety pins, locking bolts/ locking arms and safety covers are locked and in position.
- 7.3. Inspect the worksite and identify the extent of the work to be carried out and note any possible hazards or constraints. Underground cables, services etc. Check with relevant government departments on the location of these, before commencement of any work.
- 7.4. Prepare a plan as to the sequence of how the work is to be carried out safely.
- 7.5. Inspection of machine: Check hydraulic hosing and fittings, check "Skid Steer Machine" in accordance with "Operations Manual", check for restrictions on moving parts, check wear parts, pilot & cutting bits etc, check on auger pin, oil & fuel and other components as per manufactures "Operating Manual".
- 7.6. Machine is only to be used in accordance with the "Skid Steer Machines Operations Manual" E.g. Degree of elevated slope/ ground that the machine can safely be used etc.
- 7.7. Under **NO** circumstance can any person alter, tamper or modify with any "Relief Valves", "Engine Settings" or "Machine Settings".
- 7.8. Under **NO** circumstance should the "Skid Steer Machine" be lifted off the ground by the auger, while the auger is in use. This action will destabilise the "Skid Steer Machine".
- 7.9. **NO** loose clothing or body ornaments is allowed to be worn while operating the machine and unrestricted long hair must be retained so as not to get caught in the moving parts of this machine.
- 7.10. The operator must operate the machine in accordance with the manufacturers "Operation Manual".
- 7.11. The engine must be "shut down", tagged and locked and the machine be in "rest position" to perform maintenance.
- 7.12. If a safety observer/ off sider is being utilised he shall be in charge of the work site. The operator cannot move the "Skid Steer Machine" unless the operator has been directed by the safety observer/ off sider.
- 7.13. The safety observer/ off sider shall only direct the operator to move the machine when the safety observer/ off sider deems himself to be in a safe place.

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- 7.14. Only the operator can attach or detach auger assembly (quick hitch attachment) from the "Skid Steer Machine".
- 7.15. **No** person shall position themselves directly under the lift arms or auger attachment.
- 7.16. The appropriate personal protection equipment must be worn in accordance with "Workplace Occupational Health & Safety" regulations.
Hard hat, safety boots, gloves, safety glasses, UV protection, industrial work-wear etc.
- 7.17. When changing auger and pilot bits, a soft-headed hammer **MUST** be used.
- 7.18. The machine **MUST NOT** be used to lift and move people, equipment or materials.
- 7.19. All augers' connections and parts must be to the recommendations of the manufacturer or have the authority by the manufacturer for the use of other attachments.
- 7.20. All worn and replacement wear parts must be repaired or replaced in accordance to the manufacturer's recommendations.
- 7.21. Operators must use correct lifting techniques and postures for manual handling, changing of auger's, and when operating the machine. I.e. Back must always be in a straight position and bend the knees.
- 7.22. **No** person is to come within 6 metres of the operating plant without first gaining the attention of the operator and/or safety observer/ off sider and his approval acknowledged before entering the work site.
- 7.23. Any excavations or holes that are hazardous in a high traffic or public area or are to be left open for any period of time shall be barricaded and/ or clearly identified & they shall comply with the regulations and provisions of the "Work Health & Safety Act", regulations, codes of practice and advisory standards.
- 7.24. If using a safety observer/ off sider they must direct the operator of the "Skid Steer Machine" when moving within the work site.
- 7.24. The operator and/or safety observer/ off sider must keep all other persons and plant 6 metres clear of "Skid Steer Machine".
- 7.25. If one-man operation, extra care should be taken in identifying the presence of personnel and plant, before and during movement of the machine, still maintaining a 6-metre clearance.
- 7.26. When the "Skid Steer Machine" is travelling with the auger in a vertical position, the auger must be kept as to the ground as possible, under the guidance of the safety observer/ off sider.
- 7.27. When the "Skid Steer Machine" is travelling with the auger locked in a horizontal position; the tip of the auger should be marked with a high visibility device, so as to be clearly visible to all other persons on the work site. This also is under the guidance of the safety observer/ off sider.
- 7.28. On entering or egressing the work site care must be taken to avoid all other personal and plant.

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8.0. CLEANING UP OF WORKSITE

8.1. The worksite must be cleaned up as per the customer's specifications and in accordance with State and Local Government regulations.

1.0 REFERENCES

REFERENCES

1. Workplace Health and Safety Act 1995
2. Workplace Health and Safety Regulation 1995
3. Advisory Standard for Plant
4. Worksafe NOHSC: 1010 (1994)

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APPENDIX 2

STANDARD WORK PROCEDURE

DIGGA AUSTRALIA PTY LTD

AUGER ATTACHMENT

FOR

EXCAVATOR OR BACKHOE

1.0. SCOPE

- 1.1. This specification covers the sinking/ boring of holes using the "Auger" for a "Excavator or Backhoe".
- 1.2. The above may be carried out in public land, private property, construction & building sites, rural & urban areas & mine sites.
Where conditions suit the use of such a machine.

2.0. OBJECTIVES

- 2.1. The objectives of this "Standard Work Procedure" are to follow safe and approved techniques when sinking/ Augering holes.
- 2.2. It is a requirement of the Workplace Health and Safety Act 1995 that safe systems of work are employed and the contents of this "Standard Work Procedure" are a fundamental requirement of the obligations imposed under this Act.
- 2.3. The principal / operator and all relevant staff are responsible for ensuring that defined standards and work practices, described in this and any other relevant "Standard Work Procedure", are applied at the work place and that individual actions do not endanger the health and safety of themselves and others.
- 2.4. The principal and operator are responsible for ascertaining and obtaining all required permits, consents, licenses or other necessary approvals to carry out the work prior to commencement of the work. Any rules or conditions or requirements of any consent, permit or license must be complied with by the principal /operator in full.
- 2.5. At all times during the work, all property such as buildings, fences, equipment, trees or shrubs that are to be retained, must be safe guarded.

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3.0. PUBLIC RELATIONS/ DISCLAIMER

- 3.1. The principal / operator shall take due note of the possibility of adverse public comments and potential harm that can be done to the company's or the customer's image and is advised of the high priority that is placed on retaining this image.
- 3.2. Any complaints from the public are to be treated courteously and these, as well as any media enquiries shall be reported to the company and the customer's supervising officer, as soon as possible.
- 3.3. In all instances hole sinking/ Augering work may only be carried out where this work has received the required documented approvals from all state and / or local government authorities, public organisations or private owners. Particular care should be taken where overhead and underground services may be located.
- 3.4. These "Standard Work Procedures" are a guide only and it is the Principal / Operator's responsibility to ascertain and comply with detailed work procedures appropriate for each particular circumstance.
- 3.5. "Digga Australia Pty.Ltd." makes no representation or warranty whatsoever as to whether the above guidelines are comprehensive and the Principal and operator acknowledge that by the use of these guidelines they take full responsibility for any damage, consequential loss or other action or claim suffered or incurred by the Principal or Operator or any third party as a result of operating the "Excavator or Backhoe with Auger Attachment" in accordance with these guidelines.
- 3.7. The Principal and Operator free and release "Digga Australia Pty. Ltd." from all liability for any loss suffered or claim or action made against the Principal or Operator during or as a result of operating "Excavators or Backhoes with Auger Attachment" following these work procedures.

4.0. PLANT, TOOLS, EQUIPMENT AND MAINTENANCE

4.1. PLANT

- 4.1.1. In the event that the work requires the use of other plant, such as excavators, front end loaders or trucks, in a supportive role, the Principal Contractor must ensure that this supportive plant is of a standard and is operated in accordance with approved "Standard Work Procedures", government requirements, manufacturers "Operators Manuals" etc.

4.2. TOOLS AND EQUIPMENT

- 4.2.1. Appropriate safety clothing, i.e. safety helmet, safety boots etc. in accordance with the Workplace Health and Safety Act and relevant industry standard safety requirements.
- 4.2.2. Roadway warning signs in accordance with the "Queensland Transport Manual of Uniform Traffic Control Devices; Part 3-Works on Roads, or equivalent.

4.3. MAINTENANCE

- 4.3.1 All plant, tools and equipment are to be maintained in good working order in conformance with the requirements of the relevant Acts, Regulations or Ordinances.
- 4.3.2. During any periodic maintenance on the "Excavator or Backhoe with Auger Attachment" or other attachments, it is required that the "Excavator or Backhoe" be positioned for maintenance and the engine of the machine then is stopped. All controls must be tagged and locked (with the key removed) in the "rest position" and all pressure released from the system.

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- 4.3.3.** At any time that the machine is under maintenance, maintenance personnel must use a lockout system to eliminate the possibility of accidental start up. I.e.: All controls tagged and locked, with the maintenance personnel retaining the key until the maintenance period is finished.

5.0. SAFETY

- 5.1.** An "Operating Procedures Risk Assessment" in accordance with the requirements of the Workplace Health and Safety Act 1995 has been carried out.
- 5.2.** All work must be carried out in accordance with the Risk Assessment and with existing government and / or relevant industry safety requirements.
- 5.3.** All work must be carried out in accordance with the Manufacturer's "Owner's Manual"
- 5.4.** The "Excavator or Backhoe Operations Manual" must have been read and understood by the operator.
- 5.5.** The operator and safety observer/ off sider both must have read and understood the Digga Australia Pty Ltd "Operation Manual", "Risk Assessments" & "Standard Work Procedures" before any work is commenced.
- 5.6.** Any unauthorised modification carried out to the machine voids any Risk Assessment, warranty and responsibility of the manufacturers and their agents.
- 5.7.** Regardless of the type of work to be carried out and before any work is commenced, the operator must be aware and inform himself of any potential hazards that might exist and take appropriate measures to eliminate them.
- 5.8.** The operator shall especially be aware of the hazards of working near overhead or underground services and shall instruct his staff in such matters as detailed in existing industry safety specifications.
- 5.9.** When accessing the machine or refueling the machine the operator shall stop the engine and bring all hydraulic controls to the "rest position" before leaving the machine.

6.0. ACCREDITATION AND TRAINING

All operators must have read and understood the manufacturers "Operators Manual" and "Standard Work Procedures" and required to adhere too and/or perform the following: -

- 6.1** Adequate training by an approved person.
- 6.2.** Any training conducted by Digga Australia Pty Ltd or their nominated agents must be documented on completion.
- 6.3.** Any further training conducted by the owner must also be documented to identify trained operators.
- 6.4.** Any other accreditation, training and / or licensing to meet the specific requirements of any other relevant industry, e.g. unrestricted ACDC license, safety awareness for non-electrical workers, first aid and site induction courses etc.

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7.0. WORK PROCEDURES

All operators and/or safety observers/ off siders must have read and understood The "Operators Manual", "Standard Work Procedure" and performed any Required training as set out by the manufacturer before using this machine.

- 7.1. The operator of the "Excavator or Backhoe" must have all necessary tickets and competency required under the statues, regulations, codes of practice and advisory standards, to operate a "Excavator or Backhoe".
- 7.2. Before work commences, ensure that all safety pins, locking bolts/ locking arms and safety covers are locked and in position.
- 7.3. Inspect the worksite and identify the extent of the work to be carried out and note any possible hazards or constraints. Overhead/ underground cables, services etc. Check with relevant government departments on the location of these, before commencement of any work.
- 7.4. Prepare a plan as to the sequence of how the work is to be carried out safely.
- 7.5. Inspection of machine: Check hydraulic hosing and fittings, check "Excavator or Backhoe" in accordance with "Operations Manual", check for restrictions on moving parts, check wear parts, pilot & cutting bits etc, check on auger pin, oil & fuel and other components as per manufactures "Operations Manual".
- 7.6. Machine is only to be used in accordance with the "Excavator or Backhoe Operations Manual" E.g. Degree of elevated slope/ ground that the machine can safely be used etc.
- 7.7. Under **NO** circumstance can any person alter, tamper or modify with any "Relief Valves", "Engine Settings" or "Machine Settings".
- 7.8. Under **NO** circumstance should the "Excavator or Backhoe" be lifted off the ground by the auger, while the auger is in use. This action will destabilise the "Excavator or Backhoe".
- 7.9. **NO** loose clothing or body ornaments is allowed to be worn while operating the machine and unrestricted long hair must be retained so as not to get caught in the moving parts of this machine.
- 7.10. The operator must operate the machine in accordance with the manufacturers "Operation Manual".
- 7.11. The engine must be "shut down", tagged and locked and the machine be in "rest position" to perform maintenance.
- 7.12. The safety observer/ off sider is being utilised he shall be in charge of the work site. The operator cannot move the "Excavator or Backhoe" unless the operator has been directed by the safety observer/ off sider.
- 7.13. The safety observer/ off sider shall only direct the operator to move the machine when the safety observer/ off sider deems himself to be in a safe place.
- 7.14. The operator must not leave the machine while the safety observer/ off sider is changing augers or adding extension bars. The operator can only move the machine under the direction of the safety observer/ off sider.
- 7.15. **No** person shall position themselves directly under the boom or dipper arms or auger attachment.

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- 7.16. The appropriate personal protection equipment must be worn in accordance with "Workplace Occupational Health & Safety" regulations.
Hard hat, safety boots, gloves, safety glasses, UV protection, industrial work-wear etc.
- 7.17. When changing auger and pilot bits, a soft-headed hammer **MUST** be used.
- 7.18. The machine **MUST NOT** be used to lift and move people, equipment or materials.
- 7.19. All augers, connections and parts must be to the recommendations of the manufacturer or have the authority by the manufacturer for the use of other attachments.
- 7.20. All worn and replacement wear parts must be repaired or replaced in accordance to the manufacturer's recommendations.
- 7.21. Operators must use correct lifting techniques and postures for manual handling, changing of auger's, and when operating the machine. I.e. Back must always be in a straight position and bend the knees.
- 7.22. **No** person is to come within one and half times of the highest point of the "Excavator or Backhoe" while the plant is operating, without first gaining the attention of the safety observer/ off sider and his approval acknowledged before entering the work site.
- 7.23. Any excavations or holes that are hazardous in a high traffic or public area or are to be left open for any period of time shall be barricaded and/ or clearly identified or they have to comply with the regulations and provisions of the "Workplace Health & Safety Act", regulations, codes of practice and advisory standards.
- 7.24. The safety observer/ off sider must direct the operator of the "Excavator or Backhoe" when moving within the work site.
- 7.25. The safety observer/ off sider must keep all other persons and plant one and a half times, of the highest point of the "Excavator or Backhoe" , in distance, away from the machine at all times.
- 7.26. When the "Excavator or Backhoe" is travelling with the auger in a vertical position, the auger must be kept as to the ground as possible, under the guidance of the safety observer/ off sider.
- 7.27. When the "Excavator or Backhoe" is travelling with the auger locked in a horizontal position; the tip of the auger should be marked with a high visibility device, so as to be clearly visible to all other persons on the work site. This also is under the guidance of the safety observer/ off sider.
- 7.28. On entering or egressing the work site care must be taken to avoid other personal and plant.

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8.0. CLEANING UP OF WORKSITE

- 8.1.** The worksite must be cleaned up as per the customer's specifications and in accordance with State and Local Government regulations.

1.0 REFERENCES

REFERENCES

- 1.** Workplace Health and Safety Act 1995
- 2.** Workplace Health and Safety Regulation 1995
- 3.** Advisory Standard for Plant
- 4.** Worksafe NOHSC: 1010 (1994)